

GODWIN K KANGOR

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EDUCATION

Dartmouth College, Hanover, NH (*Major in Computer Science Modified with Digital Arts*) **Expected Graduation May 2027**

- **Relevant Coursework:** Introduction to Computer Science, Problem Solving via Object Oriented Programming, Computer Animation, Software Design & Implementation, 3D Modelling, Machine Language & Statistical Data, Calculus, Foundations of Applied CS, Computer Animation, Full-stack Web Development, Artificial Intelligence, UI & UX Design
- **Affiliations:** [Great Issues Scholars](#), [Black Praxis](#), Dartmouth Running Team, [The Booth](#), [DASA](#), [DSOCSS](#), [AfroEnsemble](#), [Bones Gate](#), [DDS](#), [DSKA\(President\)](#), [ColorStack](#), [CodePath](#)

Kabianga High School, Kericho, Kenya

April 2022

- Honors/awards: Top 0.1% in the Kenya Certificate of Secondary Examination (Nation-wide)

PROJECTS

[Calendai\(JavaScript,TypeScript,CSS,HTML\)](#)

- Built full-stack calendar application with 3 users, featuring automated deadline extraction from PDFs/websites, Google/Apple calendar/Canvas API, and AI assistant that improved scheduling efficiency significantly.

[Tiny Search Engine \(C\)](#)

- Developed search engine in C capable of processing 5000+ web pages, achieving 12% faster indexing than baseline implementation through memory-safe data structures and optimized algorithms.

[Kevin Bacon game\(java\)](#)

- Built graph-based solution using BFS to determine shortest actor connections, handling datasets with 100,000 nodes and achieving 50ms average query response time.

[2D Space management \(Java\)](#)

- Built spatial indexing system using Point Quadtree to handle 5000-point objects, reducing collision detection time from 120ms to 36ms in 2D space applications.

[Parts of Speech Tagger\(java\)](#)

- Implemented Hidden Markov Model with Viterbi Algorithm for POS tagging, achieving 91% accuracy on test corpus and processing 50,000+ words per second.

[Gold Nuggets Game\(C\)](#)

- Collaborated on a multiplayer treasure-hunt game with server-client architecture, implementing fog-of-war visibility, gold distribution, and efficient map data structures to support real-time play across 200+ nodes.

[Webcam-based painting program\(java\)](#)

- Built interactive painting program with webcam input, implementing stack/queue data structures and flood-fill algorithms, supporting real-time rendering at 30fps.

[Compressor program \(Java\)](#)

- Developed Huffman encoding-based compressor in Java utilizing priority queues and hash maps, reducing file sizes by 40-60% while maintaining <25ms decompression time.

[Client & Server \(Drawing Program\)\(java\)](#)

- Developed multi-user drawing program with Java client-server architecture, supporting 25+ simultaneous connections and real-time canvas updates with 99% uptime reliability.

LEADERSHIP AND SERVICE

CodeKenya, Nairobi, Kenya

December 2024-continue

Logistics Manager, Teaching Assistant

- Streamlined onboarding process for 60+ students in CodeKenya program, implementing resource allocation system that reduced setup time by 30% and improved student satisfaction scores.
- Provided technical support to 40+ bootcamp students, troubleshooting coding assignments in (Python, JavaScript, React) and achieving 25% improvement in project success rates.
- Led weekly mentoring sessions for students, guiding them through software development workflows and project reviews, with 70% of mentees successfully completing final projects.
- Partnered with instructional team to implement new teaching methods for core CS concepts, resulting in 20% increase in student pass rates for algorithms and data structures modules

Street Codes, Nairobi, Kenya

June 2022-continue

Co-founder

- Led the organization of coding boot camps aimed at empowering 50+ underprivileged local youth with essential programming skills, focusing primarily on Python.
- Designed comprehensive Python curriculum covering 12 key programming concepts, resulting in 70% of beginners demonstrating proficiency in fundamental coding skill
- Implemented a follow-up system to track participants' progress, with over 60% of attendees progressing to intermediate-level courses/projects within six months, 20% getting scholarships to Moringa Coding School

SKILLS

Technical Skills

- **Proficient:** Java, Python, JavaScript, C, React, Nextjs, TypeScript **Experience with:** Firebase, Google APIs, SQL, Mayo, Figma, Git, REST APIs